

Juan Hernández

Clinical Data & Process Optimization Analyst | Python Automation & QC

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TECHNICAL SKILLS

Programming & Tools

Python, SQL basic

Data Analysis

Pandas, NumPy, Statistical Analysis

Visualization & Dev

Matplotlib, Seaborn, Plotly,
Streamlit, Jupyter, Advanced Excel

Workflow Automation

n8n, API rest, HTTP request,
Selenium, Tesseract

DOMAIN EXPERTISE

- Clinical Data Processing & QC
- Molecular Diagnostics
- Laboratory Automation
- Workflow Optimization
- QA/QC Compliance

EDUCATION

M.Sc. Plant Biotechnology

CINVESTAV-UGA | 2024

B.Sc. Biochemical Engineering

TEC. Cd. Hidalgo | 2021

LANGUAGES

Spanish: Native

English: B2 (Technical)

EXECUTIVE SUMMARY

Data-driven professional with specialized experience in **molecular diagnostics, clinical workflows, and administrative process optimization**. Passionate about automating laboratory operations using Python to reduce manual workload and enhance data reliability. Strongly focused on quality control and assurance (QC/QA), operational efficiency, and error reduction within high-throughput clinical, administrative, and research environments.

PROFESSIONAL EXPERIENCE

Molecular Biology & Data Analyst

May 2024 – Present

Hospital MAC — Irapuato, Mexico

- High-Throughput Assays:** Processed and validated data for 600+ clinical diagnostic tests per month in a high-volume environment.
- Workflow Automation:** Automated laboratory report generation using Python, reducing manual workload by **~40%** and minimizing operational errors.
- Performance Monitoring:** Developed visualization dashboards to track operational metrics and quality control.
- Regulatory Compliance:** Ensured compliance with clinical quality standards and current regulations.
- Proposals and MVPs:** Designed a comprehensive digitization proposal for the department (MVP) with an implementation plan, presented to management.

Junior Researcher

Aug 2021 – Aug 2023

CINVESTAV — Irapuato, Mexico

- Experimental Design:** Formulated and executed high-precision molecular biology experiments for biotechnology research.
- Statistical Analysis:** Analyzed complex experimental datasets employing standard statistical methods.
- Custom Data Tools:** Developed custom Python scripts for streamlined data processing and visual analytical reports.
- Technical Documentation:** Authored technical reports and research documentation.

KEY PROJECTS

- Clinical Data Analysis Pipeline:** Python pipeline for diagnostic data incorporating statistical modeling and interactive visualizations.
- Bioinformatics Sequence Analyzer:** Automated tool for DNA and protein sequence analysis using Biopython.
- Laboratory Data Dashboard:** Streamlit web application designed for real-time automated quality control monitoring.